GEK- 65464

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

VIBRATION AMPLIFIER BOARD

MODEL NO. 3S7700PB111__

FOR

TURBINE SUPERVISORY INSTRUMENT SYSTEM

GENERAL ELECTRIC

ISSUED: October 1979

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

INTRODUCTION

Description

The model no. 3S7700PB111 -- vibration amplifier board is designed to replace the older catalog number 5473288 or 1597K50 vibration amplifier board, and is physically and electrically interchangeable with either of these older boards. The circuit board is located in the vibration power unit, (catalog number 5482034) which is part of the Turbine Supervisory Instrument Systems having catalog numbers 5482400 through 5482449.

The amount of vibration is usually indicated and/or recorded in mils peak-to-peak vibration displacement amplitude. (One mil = 0.001 inch = 0.0254 mm.)

These instructions cover only the replacement vibration amplifier board. For instructions on the vibration detectors and vibration power unit, refer to the separate instruction publications for these components. For operating instructions for all the components combined, refer to the applicable Turbine Supervisory Instrument System instructions.

Vibration Detectors

The vibration amplifier board (model number 3S7700PB111) is suitable for use with any of the General Electric vibration detectors shown in Table I, all of which have an output of 0.4 volts rms per inch per second of rms vibration velocity.

TABLE I

DETECTORS

Detector Model No. or Cat. No.	Detector Instructions
3S7700VB100--	GEK-36546
8560K12	GEK-36457
50-360---	4542K34-001
50-361---	4542K34-001
5470364	4542K11-001
9890420	GEI-36807

SPECIFICATIONS

Scale Rating

When shaft vibration detectors are used, the recorder and/or indicator is usually provided with a scale of 0-15 mils. When bracket-mounted detectors are used, a scale of 0-5 mils is generally furnished. Vibration amplifier boards and their principal characteristics are shown in Table II.

Output

The output of the vibration amplifier board is 0 to 2 milliamperes (MA). The negative side of the output is grounded within the amplifier board, and neither side may be grounded at the recorder. The recorder input resistance should not exceed 100 ohms. If an indicating instrument is used, it should be rated 0 to 2 MA, and neither terminal of the instrument may be grounded. If two indicating instruments are used, connect them in series. The resistance of the indicating instrument (or the sum of the two, if two are used) should not exceed 100 ohms.

TABLE II
VIBRATION AMPLIFIER BOARDS

Model No. *	Mils Full Scale	Low-Frequency Compensation**	Frequency of 115 V Supply Voltage	Calibrating Voltage (rms) at TP9
3S7700PB111A-	15	yes	60	0.800
3S7700PB111A-	15	no	60	0.800
3S7700PB111C-	5	no	60	0.267
3S7700PB111D-	15	yes	50	0.667
3S7700PB111E-	15	no	50	0.667
3S7700PB111F-	5	no	50	0.222

* The final letter of the model number is followed by a single numeral (factory style number). Changes in this numeral do not signify any change in above characteristics.

** Low-frequency compensation, when furnished, provides increased amplification at frequencies from 7 Hz to 15 Hz, to compensate for the fall-off of detector output voltage at these low frequencies.

Regulated D-C Power-Supply Voltages

Positive (TP6) -- +13.9 to +16.1 V
Negative (TP8) -- -13.9 to -16.1 V

RECEIVING AND HANDLING

Avoid rough handling of the circuit board. Also, avoid touching the printed circuit side of the board with your fingers, as this may leave fingerprints under humid conditions. If the circuit board is to be stored prior to installation, store it in its original protective packaging, in a place where it will not be exposed to moisture, temperature extremes or corrosive atmosphere.

INSTALLATION

To install the circuit board, proceed as follows:

1. Turn OFF the switch on the front of the vibration power unit.

2. Open the power unit drawer as far as possible, then swing the instruction card up out of the way.

3. Remove the four #6-32 circuit-board mounting screws. Disengage the old circuit board from its socket and remove the old board. If the old circuit board has a pair of inductors mounted on a bracket in the rear of the power unit, remove and discard this inductor and bracket assembly, as the inductors are not used with the new replacement circuit board.

4. Make sure the clear plastic sheet (originally furnished with the vibration power unit) is in place under the circuit board. This sheet protects the operator from accidental contact with the printed wiring on the bottom of the board, and should be used with the new replacement circuit board.

5. Insert the new replacement circuit board into its socket, making certain it is pressed all the way in.

6. Replace the four circuit board mounting screws, swing the instruction card back down into place, and close the power unit drawer.

7. Turn ON the switch on the front of the power unit.

NOTE: The model 3S7700PB111 -- replacement vibration amplifier board has three trimmer-potentiometer adjustments: P1, the CAL. ADJ.; P2, the ZERO ADJ.; and P3, the CAL. VOLTAGE ADJ. All three were correctly set at the factory, and should not be disturbed, except as described under CHECKS, ADJUSTMENTS AND TESTS.

CHECKS, ADJUSTMENTS AND TESTS

The checks, adjustments and tests described below cover only the vibration channels. For the procedure for other channels, refer to the applicable Turbine Supervisory Instrument System instructions.

1. Turn the rotary selector switch on the vibration power unit to RUN. If there is an expansion power unit, also turn its rotary selector switch to RUN. The multipoint recorder should cycle through the vibration channels and the expansion channels (if any).

2. Turn the rotary selector switch to ZERO AMPL.

   (a) The recorder should stop on one of the vibration chart sections and the pointer should indicate zero. The TEST lamp should be lit and the RUN lamp should be out.

   (b) If the recorder does not indicate zero, check the recorder zero by pressing the recorder ZERO switch. If the recorder still does not indicate zero, refer to the recorder instructions for adjustment procedure. Release the ZERO switch.

   (c) If the recorder zero is correct but a zero reading still cannot be obtained on the ZERO AMPL. tes, adjust the ZERO ADJ. (P2) on the vobration amplifier board in the power unit.

   NOTE: The ZERO ADJ. (P2) was correctly set at the factory, and ordinarily should require little or not readjustment when placing the equipment in operation. Do not disturb the setting of the CAL. VOLTAGE ADJ. (P3).

3. Turn the rotary selector switch to RUN. Allow the recorder to cycle to the other vibration chart section, and then turn the rotary selector switch to ZERO AMPL. The recorder should again indicate zero, without any further adjustment of the zero adj. (P2) on the vibration amplifier board. If it does not, recheck the recorder zero by pressing rhe ZERO switch in the recorder.

4. Turn the rotary selector switch to CAL AMPL. The recorder should indicate full scale, within $\pm 3\%$. The TEST lamp should be lit and the RUN lamp should be out. If the recorder reading is not within $\pm 3\%$ of the full scale rating of the vibration chart section, adjust the CAL. ADJ. (Pl) on the vibration amplifier board.

   NOTE: The CAL. ADJ. (Pl) was correctly set at the factory, and little or not readjustment should be required when placing the equipment into operation. Do not disturb the setting of the CAL. VOLTAGE ADJ. (P3).

PRINCIPLES OF OPERATION

Vibration Amplifier

The elementary diagram is shown in Figure 1. The vibration detector provides an alternating voltage of the same frequency as the vibration, the magnitude of the voltage being proportional to the vibration velocity. This voltage is integrated, providing an alternating voltage proportional to the vibration displacement amplitude.

Whichever detector is selected by the sampling switch in the recorder is connected to input terminal X and to ground (terminal R). The integrator consists of A1, together with an input resistance (P1 plus R1) and a feedback capacitor, C3. Trimmer potentiometer P1 provides the calibration (SPAN) adjustment.

The low-frequency compensating section consists essentially of an operational amplifier, A2, together with R5 and R6 in series, F7, C6, C7, R9, and R10, jumper "B" being omitted. This circuit provides essentially constant gain from 15 Hz up, with increasing gain as the frequency is decreased from 15 to 7 Hz. On circuit boards thast do not have low-frequency-compensation, C6, C7 and R7 are omitted and jumper "B" is included, thus providing conventional resistance feedback through R9 and R10 in series.

Capacitors C1, C5, C8, and C9 are included to remove any undesirable high frequency noise.

After amplification by A2, a peak-to-peak reading circuit is used. The principal components of this circuit are capacitors C10 and C11, diode D5, transistor Q1, and clamping diode D4 in conjunction with amplifier A3. As the alternating-voltage output from A2 passes through the negative portion of its cycle, the left side of C10 is charged to the negative peak value, while the right (+) side of C14 is clamped to +0.9V by A3 and A4. Then on the peak of the positive half cycle, the right (+) terminal of C10 rises to +0.9V plus the peak-to-peak value of the wave, thus causing emitter follower Q1 to charge C11 to the peak-to-peak value.

The ZERO ADJ. P2, is provided to set the output to 0 MA when there is no vibration input. Diodes D1, D2, and D3 compensate for changes in D5 and the base-emitter characteristic of Q1 with changes in temperature. Amplifier A4 is connected as a voltage follower, thus minimizing the amount of discharge of C11 between positive peaks at low vibration frequencies.

Power Supply

The elementary diagram is shown in Figure 2. An unregulated positive voltage (approximately +31V) is provided by the 48-volt secondary winding of transformer T1, together with rectifiers D6 and D7, resistor R21 and capacitor C13. Voltage regulator VR1, together with sensing resistors R26 and R27, provides the positive regulated voltage. This voltage is nominally +15V, but in any individual circuit board it may differ somewhat from this figure (see SPECIFICATIONS). A current-sensing resistor, R25, is included to limit the current to a safe value in case of an accidental overload or short-circuit of the regulated output.

The negative unregulated voltage (approximately -28V) is provided by rectifiers D8 and D9, resistor R28 and capacitor C16. Voltage regulator VR2, together with resistors R29 through R32 and pass-transistor Q3, provides the negative regulated voltage. This voltage is nominally -15V, but in any individual circuit board it may differ somewhat from this figure (see SPECIFICATIONS). A current-sensing resistor (R34) in conjunction with a current-limit transistor (Q2) limit the current to a safe value in case of an accidental overload or short circuit of the regulated output.

The detector continuity-checking voltage is provided by a 6.3 volt secondary winding on T1, rectifier D10 and resistor R35 and R36.

The calibrating-voltage circuit consists of a second 6.3-volt secondary winding, resistors R37, R38, and R39, and trimmer potentiometer P3. This voltage was correctly set at the factory, and ordinarily should not be changed in the field. If, for any reason, it is necessary to check or readjust the calibrating voltage, use an accurate digital voltmeter having suitable AC scales, connecting the "high" terminal of the voltmeter to TP9 and the "low" (ground) terminal to TP7. The correct values for the calibrating voltage are listed in Table II.

REPLACING CAT. NO. 5473288 OR 1597K50 VIBRATION AMPLIFIER BOARDS

Vibration amplifier boards (catalog number 5473288 and 159K50) have been superseded by vibration amplifier board (model 3S7700PB111--). The recommended replacement model numbers are listed in Table III.

TABLE III

RECOMMENDED REPLACEMENTS FOR CAT. NO. 5473288 AND 1497K50 VIBRATION AMPLIFIER BOARDS

Cat. No.	Full Scale	Old Board Low-Freq. Compensation	Power Freq. (Hz)	Notes	Recommended Replacement Board Model Number
5473288G1	15 mils	No	60		3S7700PB111B1
5473288G2	5 mils	No	60		3S7700PB111C1
5473288G3	15 mils	No	50		3S7700PB111C1
5473288G4	5 mils	No	50		3S7700PB111F1
5473288G5	----	No	60	1	Not Available
5473288G6	15 mils	No	60	2	Not Available
5473288G7	5 mils	No	60	3	Not Available
1597K50G701	15 mils	Yes	60		3S7700PB111A1
1598K50G702	5 mils	Yes	60	4	3S7700PB111C1
1598K50G703	15 mils	Yes	50		3S7700PB111D1
1598K50G704	5 mils	Yes	50	4	3S7700PB111F1
1598K50G705	15 mils	No	60		3S7700PB111B1
1598K50G706	5 mils	No	60		3S7700PB111C1
1598K50G707	15 mils	No	50		3S7700PB111E1
1598K50G708	5 mils	No	50		3S7700PB111F1
1598K50G709	0.38 MM	Yes	50		3S7700PB111D1
1598K50G710	0.125 MM	Yes	50	4	3S7700PB111F1
1598K50G711	10 mils	No	60		Not available
1598K50G712	0.38 MM	Yes	60		3S7700PB111A1

NOTES:

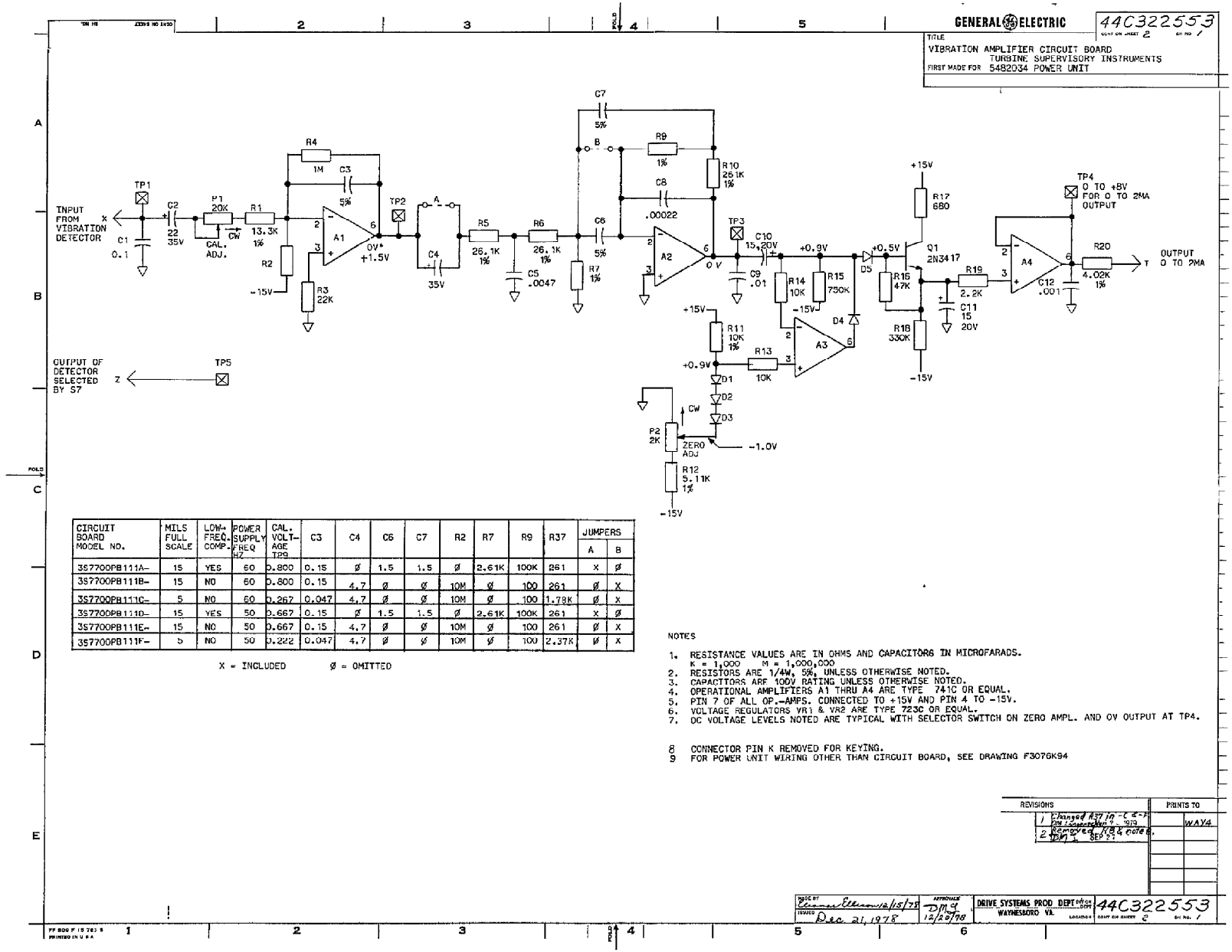
1. Special -- 1/2.5/5/10-mil scales (see GEI-77658).
2. Special -- no dc detector-check voltage. For 10-channel vibration relay (see GEI-82453).
3. Special -- for detector having an output of 0.57V/in./sec. (See GEI-88878).
4. The old board was a 5-mil or 0.125-mm full-scale board with low-frequency compensation. Low frequency compensation is no longer recommended for 5-mil and 0.125 mm boards, and is no longer available. For replacements, furnish the 5-mil board without low-frequency compensation, by ordering the recommended replacement model number listed.

REPAIR AND REPLACEMENT PARTS

If it should be necessary to check or readjust the calibrating voltage, refer to the last paragraph of the PRINCIPLES OF OPERATION section for directions.

Should a circuit board be damaged or become inoperative, it is recommended that the board be replaced. A replacement circuit board should be ordered by its model no. (see Tables II and III). Also give the model number of the turbine supervisory instrument system.

Circuit boards may, if necessary, be returened to the factory for repairs. To order a replacement circuit board, or for information on repair service, contact the nearest Installation and Service Engineering Office of General Electric Company.



CIRCUIT BOARD MODEL NO.	MILS FULL SCALE	LOW-FREQ. COMP.	POWER SUPPLY FREQ HZ	CAL. VOLTAGE TP5	C3	C4	C6	C7	R2	R7	R9	R37	JUMPERS A	JUMPERS B
3S7700PB111A-	15	YES	60	0.800	0.15	ø	1.5	1.5	ø	2.61K	100K	261	X	ø
3S7700PB111B-	15	NO	60	0.800	0.15	4.7	ø	ø	10M	ø	100	261	ø	X
3S7700PB111C-	5	NO	60	0.267	0.047	4.7	ø	ø	10M	ø	100	1.78K	ø	X
3S7700PB111D-	15	YES	50	0.667	0.15	ø	1.5	1.5	ø	2.61K	100K	261	X	ø
3S7700PB111E-	15	NO	50	0.667	0.15	4.7	ø	ø	10M	ø	100	261	ø	X
3S7700PB111F-	5	NO	50	0.222	0.047	4.7	ø	ø	10M	ø	100	2.37K	ø	X

X = INCLUDED ø = OMITTED

NOTES

1. RESISTANCE VALUES ARE IN OHMS AND CAPACITORS IN MICROFARADS. K = 1,000 M = 1,000,000
2. RESISTORS ARE 1/4W, 5% UNLESS OTHERWISE NOTED.
3. CAPACITORS ARE 100V RATING UNLESS OTHERWISE NOTED.
4. OPERATIONAL AMPLIFIERS A1 THRU A4 ARE TYPE 741C OR EQUAL.
5. PIN 7 OF ALL OP.-AMPS. CONNECTED TO +15V AND PIN 4 TO -15V.
6. VOLTAGE REGULATORS VR1 & VR2 ARE TYPE 723C OR EQUAL.
7. DC VOLTAGE LEVELS NOTED ARE TYPICAL WITH SELECTOR SWITCH ON ZERO AMPL. AND 0V OUTPUT AT TP4.
8. CONNECTOR PIN K REMOVED FOR KEYING.
9. FOR POWER UNIT WIRING OTHER THAN CIRCUIT BOARD, SEE DRAWING F3076K94

REVISIONS
1 Changed R37 to -C 2-F per Incorporation 2 - 1978
2 Removed R84 & note 8. Incorporation 3 SEP 79

PRINTS TO WAY4

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44C322553 SHT 2 SHT No. 1

Figure 1
Elementary Diagram of Vibration Amplifier Board, Amplifier Portion
(Drawing 44C322553, Sheet 1, Revision 2)

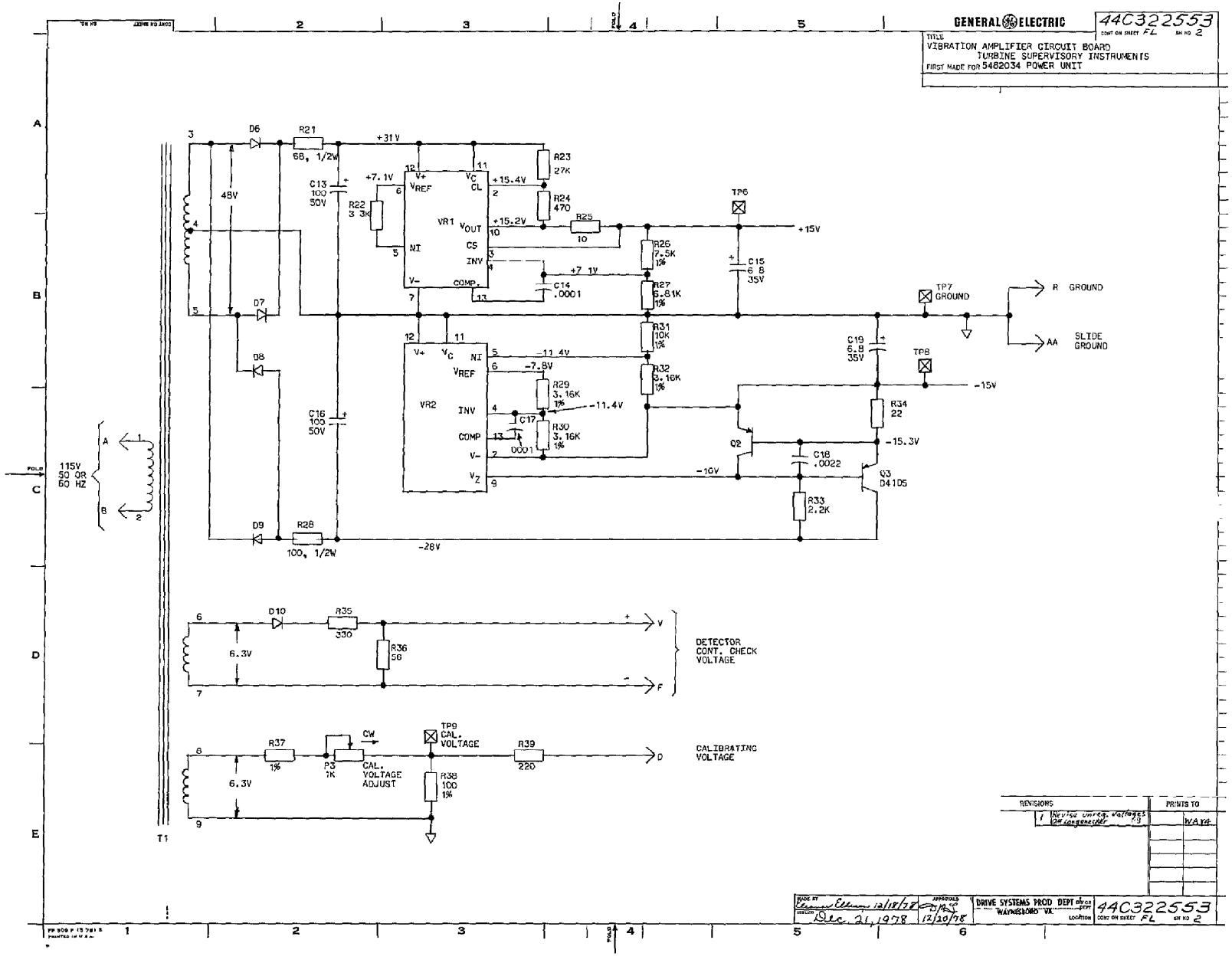


Figure 2
Elementary Diagram of Vibration Amplifier Board, Power Supply Portion
(Drawing 44C322553, Sheet 2, Revision 1)

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